

Search space: ranking pipelines

features + models + losses + ensembles



HEADROOM GATE

before any LLM spend: a hand-built candidate
must beat the noise floor by 2.5x



*no-go: stop,
keep the budget*



go

1. LLM proposes an edit

one call per iteration;
only marked code blocks change



2. Guarded evaluation

small fitness fold; leakage-safe
stats, whitelists, budgets



x 50 iterations (~\$10)

3. Scored feedback

noise-gated deltas; broken
candidates get an explanation



4. Program archive

Pareto set over relevance,
conversion, revenue



selected programs

TRANSFER AUDIT

after the search: re-score every selected program
on 60k held-out queries the loop never saw



**only gains that survive the audit
count as discoveries**